Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The elements with zero electron affinity are

(A) Boron and Carbon

(B) Beryllium and Helium

(C) Lithium and Sodium

(D) Fluorine and Chlorine

Q.2

Consider the following ionization enthalpies of two elements 'A' and 'B'.

\begin{array}{lcll}
\hline & \text{Element} & \text{Ionization enthalpy } (\mathrm{kJ} / \mathrm{mol}) \\
& 1 \mathrm{st} & 2 \mathrm{nd} & 3 \mathrm{rd} \\
\mathrm{A} & 899 & 1757 & 14847 \\
\mathrm{~B} & 737 & 1450 & 7731 \\
\hline
\end{array}

Which of the following statements is correct?

(A) Both 'A' and 'B' belong to group-1 where 'B' comes below 'A'.

(B) Both 'A' and 'B' belong to group-1 where 'A' comes below 'B'.

(C) Both 'A' and 'B' belong to group-2 where 'B' comes below 'A'.

(D) Both 'A' and 'B' belong to group-2 where 'A' comes below 'B'.

Q.3

Consider the following changes :

(1) $\mathrm{M(s)} \longrightarrow \mathrm{M(g)}$

(2) $\mathrm{M(s)} \longrightarrow \mathrm{M^{2+}(g) + 2e^-}$

(3) $\mathrm{M(g)} \longrightarrow \mathrm{M^{+}(g) + e^-}$

(4) $\mathrm{M^{+}(g)} \longrightarrow \mathrm{M^{2+}(g) + e^-}$

(5) $\mathrm{M(g)} \longrightarrow \mathrm{M^{2+}(g) + 2e^-}$

The second ionization energy of M could be calculated from the energy values associated with:

(A) $1 + 3 + 4$

(B) $2 - 1 + 3$

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(C) $1+5$

(D) $5-3$

Q.4

The electron affinity of chlorine is $3.7 \mathrm{eV} .1 \mathrm{~g}$ of chlorine is completely converted to Cl^{-} ion in a gaseous state. $\left(1 \mathrm{eV}=23.06 \mathrm{kcalmol}^{-1}\right)$
Energy released in the process is

(A) $4.8 \mathrm{kcal}$

(B) $7.2 \mathrm{kcal}$

(C) $8.2 \mathrm{kcal}$

(D) $2.4 \mathrm{kcal}$

Q.5

The electronegativity follows the order

(A) $\mathrm{F}>\mathrm{O}>\mathrm{Cl}>\mathrm{Br}$

(B) $\mathrm{F}>\mathrm{Cl}>\mathrm{Br}>\mathrm{O}$

(C) $\mathrm{O}>\mathrm{F}>\mathrm{Cl}>\mathrm{Br}$

(D) $\mathrm{Cl}>\mathrm{F}>\mathrm{O}>\mathrm{Br}$

Q.6

Electron affinity is positive for

(A) $\mathrm{O}(\mathrm{g})+\mathrm{e}^{-} \rightarrow \mathrm{O}^{-}(\mathrm{g})$

(B) $\mathrm{S}(\mathrm{g})+\mathrm{e}^{-} \rightarrow \mathrm{S}^{-}(\mathrm{g})$

(C) $\mathrm{O}^{+}(\mathrm{g})+\mathrm{e}^{-} \rightarrow \mathrm{O}(\mathrm{g})$

(D) $\mathrm{O}^{-}(\mathrm{g})+\mathrm{e}^{-} \rightarrow \mathrm{O}^{2-}(\mathrm{g})$

Q.7

The following acids have been arranged in the order of decreasing acid strength Identify the correct order (I) ClOH (II) BrOH (III) IOH

(A) $\mathrm{I}>\mathrm{II}>\mathrm{III}$

(B) $\mathrm{II}>\mathrm{I}>\mathrm{II}$

(C) $\mathrm{III}>\mathrm{II}>\mathrm{I}$

(D) $\mathrm{I}>\mathrm{III}>\mathrm{II}$

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Q.8

The correct increasing order of extent of hydrolysis is

(A) $CCl_4 < MgCl_2 < AlCl_3 < SiCl_4 < PCl_5$

(B) $CCl_4 < AlCl_3 < MgCl_2 < PCl_5 < SiCl_4$

(C) $CCl_4 < SiCl_4 < PCl_5 < AlCl_3 < MgCl_2$

(D) $CCl_4 < PCl_5 < SiCl_4 < AlCl_3 < MgCl_2$

Q.9

Sequence of acidic character is

(A) $N_2O_5 > SO_2 > CO > CO_2$

(B) $N_2O_5 > SO_2 > CO_2 > CO$

(C) $SO_2 > CO_2 > CO > N_2O_5$

(D) $SO_2 > N_2O_5 > CO > CO_2$

Q.10

In which of the following arrangements, the sequence is not strictly according to the property written against it?

(A) $HF < HCl < HBr, HI$: increasing acid strength

(B) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

(C) $B < C < O < N$: increasing first ionization enthalpy

(D) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power

Q.11

Among Al_2O_3, SiO_2, P_2O_3 and SO_2 the correct order of acid strength is

(A) $Al_2O_3 < SiO_2 < SO_2 < P_2O_3$

(B) $SiO_2 < SO_2 < Al_2O_3 < P_2O_3$

(C) $SO_2 < P_2O_3 < SiO_2 < Al_2O_3$

(D) $Al_2O_3 < SiO_2 < P_2O_3 < SO_2$

Q.12

In the long form of the periodic table, the valence shell electronic configuration of $5s^2 5p^4$ corresponds to the element present in :

(A) Group 16 and period 6

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(B) Group 17 and period 6

(C) Group 16 and period 5

(D) Group 17 and period 5

Q.13

Which pair of elements belongs to same group

(A) Elements with atomic no. 17 and 38

(B) Elements with atomic no. 20 and 40

(C) Elements with atomic no. 17 and 53

(D) Elements with atomic no. 11 and 33

Q.14

The electronic configuration of an element is $1\ s^2, 2\ s^2, 2p^6, 3\ s^2, 3p^3$. Which is the atomic number of the element which is just below the above element in the periodic table?

(A) 34

(B) 49

(C) 33

(D) 31

Q.15

An element has a configuration: $[Ar]\ 3d^3 4s^2$, predict the group or period of the element in the periodic table.

(A) Group 5

(B) Group 5 and Period 4

(C) Group 4

(D) Period 4

Q.16

Which one of the following ionic species has the greatest proton affinity to form stable compound?

(A) Cl^-

(B) F^-

(C) I^-

(D) Br^-

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Q.17

Which of the following statements is/are wrongl

(A) Van der Waal's radius of iodine is more than its covalent radius

(B) All isoelectronic ions belong to same period of the periodic table

(C) $\mathbb{E}_1$ of N is higher than that of O while $\mathbb{E}_2$ of O is higher than that of N

(D) The electron gain enthalpy of N is zero while that of P is $74.3 \mathrm{KJmol}^{-1}$

Q.18

The ionic size decreases in the order

(A) $\mathrm{K}^{+} > \mathrm{S}^{2-} > \mathrm{Sc}^{3+} > \mathrm{V}^{5+} > \mathrm{Mn}^{7+}$

(B) $\mathrm{S}^{2-} > \mathrm{K}^{+} > \mathrm{Sc}^{3+} > \mathrm{V}^{5+} > \mathrm{Mn}^{7+}$

(C) $\mathrm{Mn}^{7+} > \mathrm{V}^{5+} > \mathrm{Sc}^{3+} > \mathrm{K}^{+} > \mathrm{S}^{2-}$

(D) $\mathrm{Mn}^{7+} > \mathrm{V}^{5+} > \mathrm{Sc}^{3+} > \mathrm{S}^{2-} > \mathrm{K}^{+}$

Q.19

In which of the following arrangements, the sequence is not strictly observed according to the property written against it?

(A) $\mathrm{NH_3} < \mathrm{PH_3} < \mathrm{AsH_3} < \mathrm{SbH_3}$: increasing basic strength

(B) $\mathrm{B} < \mathrm{C} < \mathrm{O} < \mathrm{N}$: increasing first ionization enthalpy

(C) $\mathrm{CO_2} < \mathrm{SiO_2} < \mathrm{SnO_2} < \mathrm{PbO_2}$: increasing oxidising power

(D) $\mathrm{HF} < \mathrm{HCl} < \mathrm{HBr} < \mathrm{HI}$: increasing acid strength

Q.20

If electronegativity of A, B, C and D are 1.0, 1.2, 2.5 and 2.8, then the most basic compound from the following is..

(A) AOH

(B) BOH

(C) COH

(D) DOH

Answers & Solutions

Q.1 Answer:

Beryllium and Helium

Solution:

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Fully filled electronic configuration.

Q.2 Answer:

Both 'A' and 'B' belong to group-2 where 'B' comes below 'A'.

Solution:

Generally, the ionization enthalpies or energy increases from left to right in a period and decreases from top to bottom in a group. Several factor such as atomic radius, nuclear charge, shielding effect are responsible for change of ionization enthalpies.

Here, Ist ionization enthalpy of A and B is greater than group I $\left(\mathrm{Li}=520 \mathrm{kJmol}^{-1}\right.$ to $\mathrm{Cs}=$ $\left.374 \mathrm{kJmol}^{-1}\right)$, which means element A and B belong to group -2 and all three given ionization enthalpy values are less for element B means B will come below

Q.3 Answer:

$5-3$

Solution:

Second ionization energy is amount of energy required to take out an electron from the monopositive cation.

Q.4 Answer:

$2.4 \mathrm{kcal}$

Solution:

Number of moles $=\frac{1}{35.5}$ **Given,** $1 \mathrm{eV}=23.06 \mathrm{kcalmol}^{-1}$

$3.7 \mathrm{eV}=3.7 \times 23.06 \mathrm{kcalmol}^{-1}$

i.e. 1 mole realease energy $=3.7 \times 23.06 \mathrm{kcal}$

$\therefore$ **Energy released** $=\frac{1}{35.5} \times 3.7 \times 23.06 \mathrm{kcal}=2.4 \mathrm{kcal}$

Q.5 Answer:

$\mathrm{F}>\mathrm{O}>\mathrm{Cl}>\mathrm{Br}$

Solution:

F **and** O **belong to 2 nd period whereas** Cl **and** Br **belong to** $3 \mathrm{rd}$ **and 4 th periods respectively. Hence the sequence of the E.N. is** $\mathrm{F}>\mathrm{O}>\mathrm{Cl}>\mathrm{Br}$

Q.6 Answer:

$\mathrm{O}^{-}(\mathrm{g})+\mathrm{e}^{-} \rightarrow \mathrm{O}^{2-}(\mathrm{g})$

Solution:

O^{-} **ion repel the incoming electron thus energy is required to add incoming electron.**

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Q.7 Answer:

$I > II > III$

Solution:

The acidic character follows the order: HOCl > HOBr > HOI on the basis of electronegativity of halogen

Q.8 Answer:

$CCl_4 < MgCl_2 < AlCl_3 < SiCl_4 < PCl_5$

Solution:

Aqueous hydrolysis of CCl_4 is not possible due to absence of vacant orbital on carbon, while hydrolysis of $MgCl_2, AlCl_3, SiCl_4$ and PCl_5 is possible. In covalent halides, hydrolysis occurs as a result of coordination of water molecule to the less electronegative element.

Q.9 Answer:

$N_2O_5 > SO_2 > CO_2 > CO$

Solution:

The acidic character of non metal oxides increases across a period from left to right and decreases down a group. So, acidic character will follow the order:
oxide of nitrogen > oxides of sulfur > oxides of carbon. Among oxides of carbon acidic character increases with
the oxidation number of carbon. So, $^{+4}CO_2$ is more acidic than CO. Hence the sequence of acidic character is $N_2O_5 > SO_2 > CO_2 > CO$

Q.10 Answer:

$NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

Solution:

In hydrides of 15 th group elements, basic character decreases on descending the group i.e.
$NH_3 > PH_3 > AsH_3 > SbH_3$

Q.11 Answer:

$Al_2O_3 < SiO_2 < P_2O_3 < SO_2$

Solution:

As the size increases the basic nature of oxides changes to acidic nature i.e., acidic nature increases.

$$\underset{\text{Acidic}}{SO_2} > \underset{\text{Weak acidic}}{P_2O_3} > \underset{\text{Amphoteric}}{SiO_2} > Al_2O_3$$

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SO_2 and P_2O_3 are acidic as their corresponding acids H_2SO_3 and H_3PO_3 are strong acids.

Q.12 Answer:

Group 16 and period 5

Solution:

Tellurium (Te) has $5s^2 5p^4$ valence shell configuration. It belongs to group 16 and present in period 5 of the periodic table.

Q.13 Answer:

Elements with atomic no. 17 and 53

Solution:

Atomic no. $17(\mathrm{Cl})$ and $53(\mathrm{I})$ are present in the same group

Q.14 Answer:

33

Solution:

The atomic number of the above element is $2+2+6+2+3=15$. The atomic number of the element below the given element is $15+18=33$.

Q.15 Answer:

Group 5 and Period 4

Solution:

The electronic configuration is given as $\left[\mathrm{Ar}\right]\ 3\mathrm{d}^3 4\mathrm{s}^2$. In this electronic configuration sum of the number of valence electrons is equal to the group number and the highest value of principal quantum number is equal to period number. Its atomic number is 18(Ar)+5(valence electrons)=23. Hence, It is Vanadium element. It is located in 4th period and 5th group of the periodic table.

Q.16 Answer:

F^-

Solution:

Proton affinity decreases in moving across the period from left to right due to increase in charge, within a group the proton affinities decreases from top to bottom.

Q.17 Answer:

All isoelectronic ions belong to same period of the periodic table

Solution:

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In the isoelectronic series all isoelectronic anions belong to the same period and cations to the next period.

Q.18 Answer:

$S^{2-} > K^{+} > Sc^{3+} > V^{5+} > Mn^{7+}$

Solution:

The given species are isoelectronic. The size of these species decreases with increase in the positive charge.

Q.19 Answer:

$NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

Solution:

The decrease in electronegativity and increase in size of element cause the decrease in tendency to accept proton. Thus, the correct order of increasing basic strength is

$SbH_3 < AsH_3 < PH_3 < NH_3$

Q.20 Answer:

AOH

Solution:

For XOH**, Electronegativity of** X ↓ , ΔEN **of** XO ↑ **, Basic strength** ↑

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